

Course Project Details

Term 7 (Sept)-2025

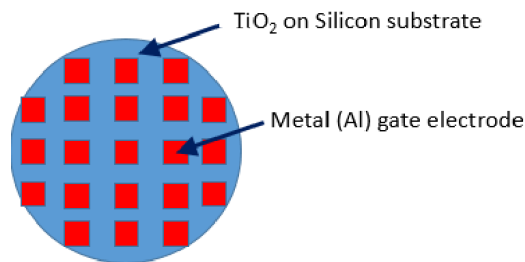
01.119/99.503: Nanoelectronics and Technology

1. Presentation

- On Dec 109 (3.300-6PM) (Week 13 Class 1)
- 30 minutes per group
- PPT presentation (No need to write report)
- Team members should take turn and all members need to present some part of the presentation
- Email PPT file in pdf format after the presentation

2. Presentation contents

- Discussion on fabrication



- Ideal MOS C-V characteristics and simulation results on C-V
- Shift in C-V characteristics from fabricated (real) samples and extraction of different parameters of the capacitor
- Find the breakdown voltage of TiO_2
- I_g –time characteristics for soft breakdown and hard breakdown
- Comparison with SiO_2 dielectric (Data will be provided)
- Analyse any other characteristics of the MOS capacitor
- Any correlation of experimental results to simulation results
- Summarize the results

Week 11 class 2 and Week 12 class 1 -Electrical characterization sessions.